

MACHUPALLE SOMA SEKHARA REDDY

DATA ANALYST

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PROFESSIONAL SUMMARY

Detail-oriented **Data Analyst and GenAI professional** with experience in insurance and pension domain projects, specializing in **Python, SQL, Power BI, Excel, LLMs, and Generative AI**. Experienced in building AI-driven solutions for document classification, information extraction, audio transcription, checklist evaluation, and automated scoring. Hands-on experience in data analysis and visualization through personal projects involving **300K+ transaction records, KPI development, dashboard creation, trend analysis, and business performance reporting**. Skilled in transforming unstructured data into structured insights and building interactive Power BI dashboards using SQL, Excel, and data analysis techniques to support data-driven business decision-making.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, Pandas, NumPy, Statistical Analysis, Data Cleaning, and Data Transformation.
- **Database & Querying:** SQL, relational databases, complex queries, joins, aggregations, subqueries, and data manipulation.
- **Data Visualization & Reporting:** Microsoft Excel and Power BI, including dashboards, reports, KPIs, and data-driven insights.
- **ETL & Data Engineering:** ETL processes, data extraction, transformation, validation, and preparation for analytics and reporting.
- **AI & Application Development:** Flask API development, Large Language Models (LLMs), Generative AI (GenAI), Prompting, and AI-powered application integration.

PROFESSIONAL EXPERIENCE

GEN AI Python Developer – TCS

Project: NYL (New York Life Insurance)

- Developed an AI-driven workflow to transcribe insurance-related audio recordings into text, creating structured input for downstream quality and compliance analysis.
- Designed a **Checklist Agent** using LLMs and prompt engineering to evaluate transcribed conversations against predefined questions and generate Yes/No/NA responses.
- Developed prompt-based evaluation logic for question-level assessment, ensuring checklist outputs were generated consistently from the transcription context.
- Built a **Scoring Agent** that consumes checklist results and calculates an overall score based on predefined evaluation criteria and prompts.
- Integrated **Python, LLMs, GenAI, prompt engineering, and API-based workflows** to automate the end-to-end transcription, checklist evaluation, and scoring process.

Project: Athora Pension Implementation

- Developed an AI-driven document processing solution for Athora pension and insurance documents, automating document classification using LLMs and prompt engineering.
- Designed and optimized prompts to identify relevant document types and **extract key-value fields** from PDF and image-based documents.
- Transformed unstructured document information into structured JSON outputs for downstream processing.
- Built and integrated Python and Flask APIs to support document processing, LLM-based extraction, and structured response generation.
- Applied GenAI, LLMs, Python, and data analysis techniques to improve automation and reduce manual effort in pension document processing.

PERSONAL PROJECTS

PhonePe Transactions Analytics Dashboard

Technologies: SQL, Power BI, Excel, Data Analysis

- Developed an interactive Power BI dashboard to analyze approximately 300K transactions with a total transaction value of 3.47B.
- Created KPIs for transaction volume, total amount, average transaction value, 96% success rate, and failed transaction analysis.
- Analyzed transaction performance by service, service type, payment status, month, and failure reason to identify key trends.
- Implemented interactive filters for month, service, payment status, and failure reason to enable detailed business analysis.

Swiggy Sales & Restaurant Analytics Dashboard

Technologies: Power BI, SQL, Excel, Data Visualization

- Developed an interactive Power BI dashboard to analyze sales, orders, ratings, order value, restaurants, dishes, and city-level performance.
- Created KPIs for total sales, average rating, average order value, total orders, and number of dishes.
- Analyzed monthly and daily sales trends to identify high-performing periods and understand business performance.
- Built visualizations for top restaurants, top dishes, locations, prices, and ratings with interactive city-level filtering.

EDUCATION

B.Tech CSE | Vel Tech Rangarajan Dr. Sagunthala R&D; Institute of Science and Technology, Chennai, Tamil Nadu | CGPA: 8.55

Intermediate MPC | NRI Junior College, Tirupathi, Andhra Pradesh | CGPA: 9.25

SSC | Pavan Concept School, Kadapa, Andhra Pradesh | CGPA: 9.7